

FEASIBILITY & VIABILITY

TECHNICAL • ECONOMIC • MARKET • SCALE

Why MarketLink can work in the real world — and how we control the risks.

01 FEASIBILITY + VIABILITY

Buildable today. Sustainable as adoption grows.

Technical Feasibility

Proven stack: Kotlin, Java/Spring Boot, PostgreSQL, MongoDB, Python/FastAPI.
Government mandi data supports current + historical intelligence.
AI/ML is a separate service; Gemini + Exotel enable voice.
Modular services can be upgraded independently.

Economic Feasibility

Cloud/API infrastructure - no specialized hardware.
Government data reduces dependence on costly proprietary datasets.
Pay-as-you-scale deployment.
Core marketplace still works if AI is temporarily unavailable.

Operational + Scale

One backend = single source of truth for users, lots, offers and transactions.
REST APIs support mobile, voice and future apps.
Database/service layers scale independently.
Marketplace operations stay separate from the AI layer.

Market + Long-Term Viability

Farmers gain buyer discovery, offers and market information.
Buyers get agricultural lots + competitive offers.
Start with focused crops/markets; expand as participation grows.
Voice + mobile access lowers the smartphone barrier.
Low-cost start + scalable infrastructure supports sustainable growth.

02 POTENTIAL CHALLENGES

The main failure modes judges should know we have planned for

MARKET DATA

API downtime; stale / incomplete mandi data

AI PREDICTION

Market volatility may reduce accuracy

VOICE

Noise, accents, regional languages, ambiguity

CONNECTIVITY

Limited or unstable internet access

ADOPTION

Too few buyers can reduce market liquidity

SECURITY

Unauthorized access to farmer / buyer / transaction data

SCALABILITY

More users, bids & real-time operations increase backend load

03 MITIGATION STRATEGIES

Concrete controls that keep the marketplace reliable, safe and usable

Data reliability

Freshness checks + validation + live/cache separation + warning/fallback.

AI quality gate

Model evaluation → quality gate → confidence; weak models are not shown as confident recommendations.

Voice + access

Exotel → Gemini → structured intent → Spring Boot → response. Clarify ambiguity; confirm sensitive actions. Voice supports smartphone-limited users; cache/fallback helps continuity.

Adoption strategy

Start with focused crops/markets; enable buyer discovery + competitive offers; expand coverage as participation grows.

Security + scale

JWT + role-based access + ownership checks; AI cannot modify marketplace data. Modular services, REST APIs, separate AI inference, PostgreSQL for transactions, MongoDB for images, replaceable adapters.

Core continuity

Critical operations remain backend-controlled; marketplace stays usable when AI/external services fail.

BUILT FOR REAL-WORLD CONDITIONS

Reliable Data • Controlled AI • Secure Transactions • Multiple Access Channels
Failure of one external component does not stop the core farmer-buyer marketplace.